Data description: making choices

Presentation Outline

Data Description

These notes assume familiarity with basic descriptive statistics:

- location: mean, median
- spread: variance, sd, coefficient of variation (Cv), se
- shape: skewness and kurtosis
- If not familiar, review material on basic tools
- These notes describe less familiar aspects of data description
- Context for this segment:
  - Statistical analysis requires making choices
  - What are some of those choices?
  - and how do you make that choice?

Should you report mean or median?

- Relationship between mean, median and skewness
- Pictures

What the pictures show

Both show data from a simple random sample of one population
When population is symmetric:

- Mean and median are the same
- Sample average and sample median are estimates of the same parameter
- Median is robust to outliers
- Why is the mean the usual choice?
- When population is a normal distribution,
  - Relative efficiency of mean is 1.57, $\sqrt{1.57} = 1.25$
  - Uncertainty of mean from 40 observations same as that for median of 50 pbs.

What the pictures show

When population is positively skewed

- Mean > Median
- How much larger? Depends on the skewness
- Pictures

What the pictures show

So should you report mean or median?

- Frequent "knee-jerk" advice: use median for skewed data
- Better: what population quantity do you want to describe?
  - Typical value: use median
  - Total: use mean
- Examples:
  - Summer intern story
  - Oil pollution story

Should you report var, sd or cv?

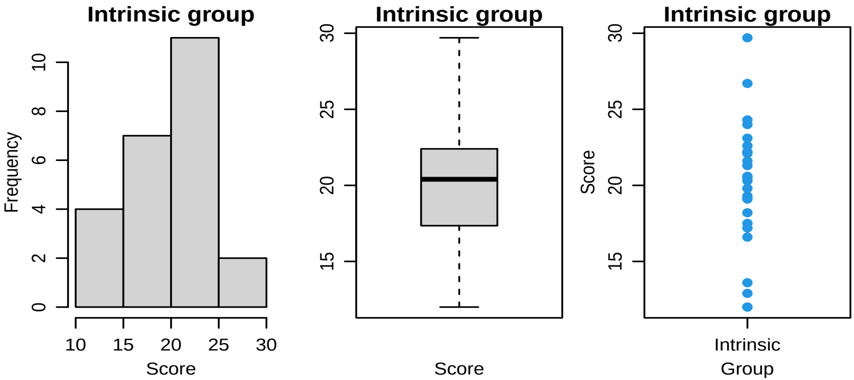
- Depends on how you want to describe the variability
- var and sd: absolute variability
  - Variance has squared units
    - Harder to interpret
    - Almost always used as an intermediate quantity
  - sd has same units as the response
- cv (coefficient of variation): variability as a proportion of the mean
  - cv = sd / mean when describing spread in observations
  - most frequently used when you expect higher mean to be associated with higher sd
  - unitless: Measurement in gm or kg have same cv
  - Some fields have "standards" for cv
  - Analytical chemistry: variability among replicate measurements
    $<$ 10% excellent, 10% - 20% good, 20% - 30% acceptable, $>$30% poor / unacceptable

Should I use a histogram, a box plot, or a dot plot of data?

Goals: A picture that shows characteristics of a group of data

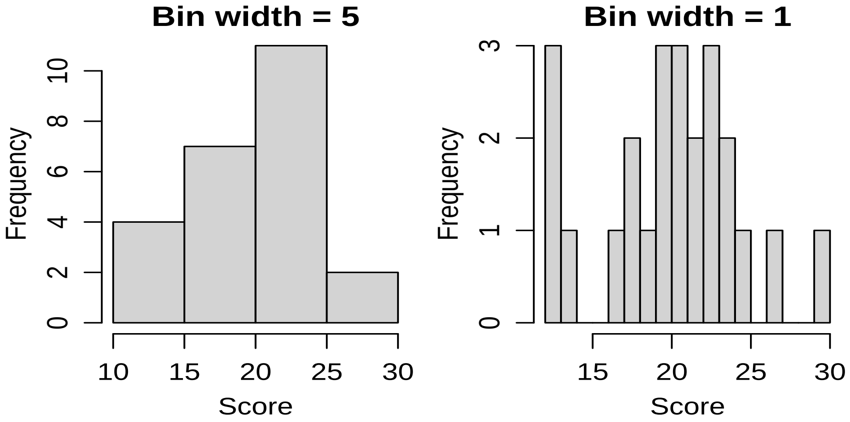
And can be used to compare two or more groups

Illustrate using data from the creativity study (Sleuth Chapter 1 case study)



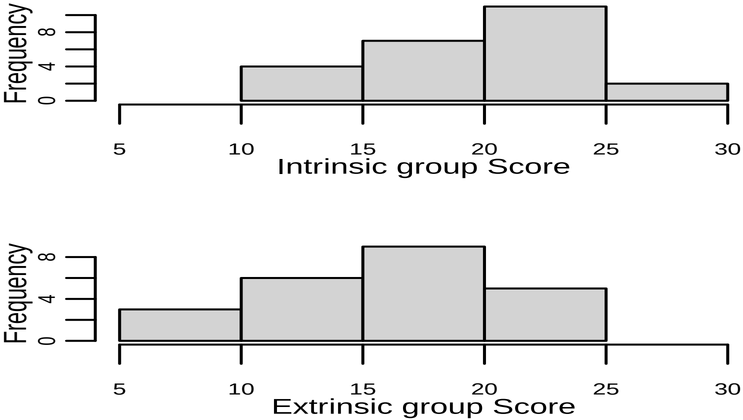
Histogram

- Requires binning the data - can change plot appearance



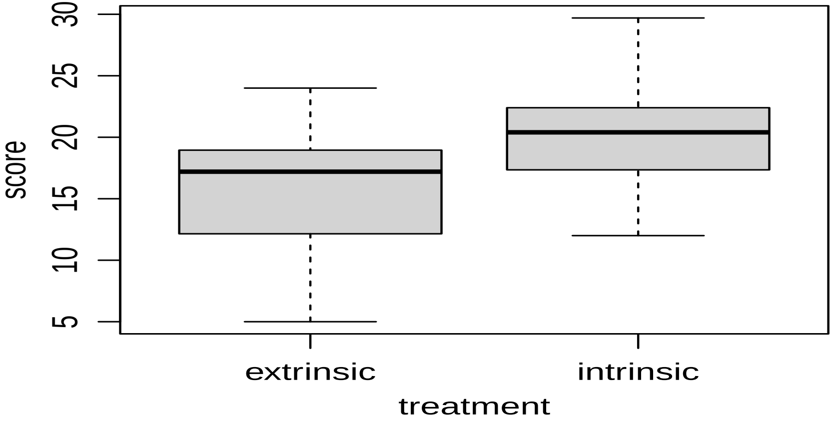
Histogram

- And gets bulky when multiple groups



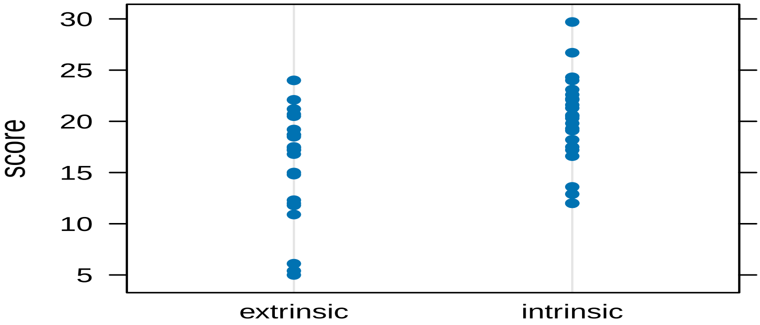
Box plot

- Very compact display, even for multiple groups



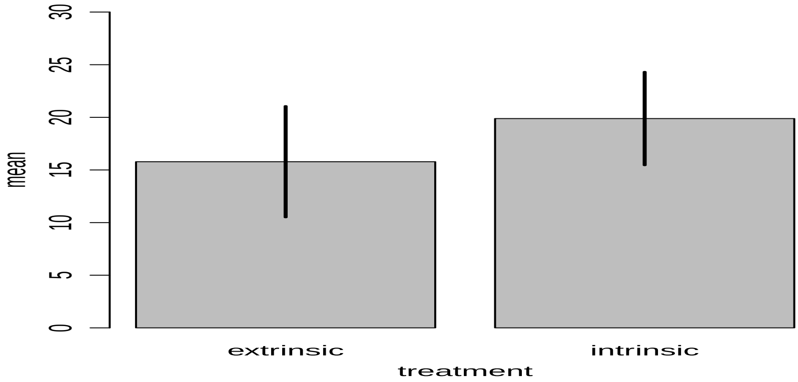
Dot plot

- When few observations, why not just show all the data?
- How few is few? Depends on the amount of overplotting



Bar graph of means

- Very popular, but poor use of ink and space
- Shows just the means, sometimes means $\pm$ sd (when describing data)



Important points:

- If any concepts are fuzzy, review the relevant module 0 material
- Should you report mean or median?
  - What is the goal? Typical value or a total
  - Doesn't matter if population has a symmetric distribution
- Should you report var, sd, or cv?
  - Is absolute variability or relative variable more relevant?
- How should you show a picture of the data?
  - Relatively few observations: dot plot
  - Otherwise: box plots